

Ayesha Binte Mostofa

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RESEARCH INTEREST

I am interested in the intersection of distributed systems, large language models, and usable security and privacy. This is related to the following areas: 1) Understanding security and privacy challenges in large-scale systems and developing scalable methods to manage them. 2) Designing user-centered security mechanisms that help people make safer decisions while interacting with online platforms.

Keywords: Security and Privacy on Cloud and Network Systems, Usable Security, LLMs

EDUCATION

University of Massachusetts Amherst (UMass Amherst)

MS/Ph.D. in Computer Science

Sep 2025 – Present

Amherst, MA, USA

Advisor : [Dr. Pubali Datta](#)

Bangladesh University of Engineering and Technology (BUET)

B.Sc.Engg. in Computer Science and Engineering

April 2019 – July 2024

Dhaka, Bangladesh

Thesis Advisor : [Dr. Anindya Iqbal](#)

RESEARCH EXPERIENCE

Security Research Group of Dr. Pubali Datta, UMass Amherst | 2025-Present

- Leading two first-author research projects on systems security and usable security.

BUET Undergrad Research | 2023-2025

- Designed prompts incorporating static program metadata (function call graphs) and natural language summaries; applied QLoRA fine-tuning to improve code review comment and code refinement generation.

Advisor: Dr. Anindya Iqbal, Dr. Toufique Ahmed

- Detected benign and malicious attack behaviors by analyzing raw log data from DARPA OPTC and DARPA TC E3 datasets using PostgreSQL and provenance graphs.

Advisor: Dr. Shohrab Hossain, Dr. Shahrear Iqbal

PUBLICATIONS

1. Quantifying Risk Perception and Scam Response Among International and Domestic U.S. University Students.

Alexandra Xinran Li, Elijah Bouma-Sims, Lily Klucinec, Ray Liu, **Ayesha Binte Mostofa**, Arjun Arunasalam, Lorrie Faith Cranor, Pubali Datta, Lucy Simko, Karen Sowon. *Symposium on Usable Privacy and Security (SOUPS)*, 2026.

2. Advancing Code Review and Code Refinement Automation Using LLMs. **Ayesha Binte Mostofa***, Sk. Sabit Bin Mosaddek*, Asif Haider*, Toufique Ahmed, Anindya Iqbal. [arXiv Preprint](#), 2024. (*Equal contribution)

POSTERS

1. Understanding Synthetic Politics on Reddit: Spread and Community Reactions to Political Deepfakes. **Ayesha Binte Mostofa** and Pubali Datta. *USENIX Security '26 Poster*, 2026.

2. 10th NSysS Research Poster Presentation, 2023 [Poster]

TALKS & SERVICE

Short Talk | Annual Boston Security Usable Research Day, 2026 at Tufts University

External Reviewer | **IEEE TDSC 2026**

Poster Presentation & Volunteer | New England Security Day 2026, UMass Amherst

IEEE Computer Society BUET Student Branch Chapter, 2021 – 2023

- Publicity Committee Coordinator (2022 – 2023); Media Committee Executive (2021 – 2022)

WORK EXPERIENCE

Research Assistant, UMass Amherst	Summer 2026, Fall 2026
Teaching Assistant, UMass Amherst • Introduction to Computation (COMPSCI 250) • Programming Methodology (COMPSCI 220)	Fall 2025, Spring 2026
Full-time Lecturer <i>Department of CSE, Presidency University; [Profile]</i> <i>Department of CSE, Canadian University of Bangladesh</i>	Dhaka, Bangladesh January 2025 – July 2025 July 2024 – January 2025
Machine Learning Intern (Part-time) <i>Red.Digital Limited</i>	May 2023 – June 2023 <i>Remote</i>

HONORS & AWARDS

USENIX Security Symposium Travel Grant, \$825 Award, 2026

GREPSEC VII Workshop Participant, Selected as 1 of 48 participants internationally; \$1175 Travel Grant Award (co-located with USENIX Security '26)

David W. Stemple scholarship in computing 2026, \$2300 Award, UMass Amherst

RISE-BUET Internal Student Research Grant for Undergraduate Thesis, 2023

Dean's List Award, BUET, Level 3 & Level 4, 2023–2024

Deep Learning Enigma 1.0 Finalists, 2024, Object Detection Contest for Autonomous Vehicles

Merit Scholarships, 2017–2024, granted for being top 1% student in HSC and SSC level

ACADEMIC PROJECTS

LLMComp: Prompt Compression for Robust Defense Against LLM Jailbreaks — 2025	<i>LLM, Few-Shot, Jailbreak</i>
IaC-Eval Replication and Extension Study — 2025	<i>IAC, LLM, Few-shot, CoT</i>
Hardware : ALU, FPA, MIPS Implementation — 2022	<i>C++, Logisim, Assembly</i> Source Code
MISP Tool, Cryptography and Malware Analysis — 2023	<i>Python, Docker, Azure Cloud</i> Source Code — Youtube Demo
C Compiler — 2022	<i>Lex, Yacc, Assembly, C</i> Source Code
Implementing functionalities of OS — 2023	<i>C, Bash</i> Source Code
TCP CUBIC-FIT, Packet Tracer and Network Simulator — 2022	<i>Java, NS2</i> Source Code

TECHNICAL SKILLS

Languages	: C/C++, Python, Java, Javascript, Assembly, Bison/Flex, Bash, MySQL, LaTeX
Frameworks	: React.js, Node.js, SpringBoot, Oracle, PostgreSQL, Docker, NS2, xv6, Git, Wireshark
Libraries	: NumPy, Keras, Matplotlib, OpenCV, OpenGL, Pandas, Scikit Learn
Platform	: Linux, Windows, MacOS

REFERENCE

Dr. Pubali Datta <i>Assistant Professor, Computer Science</i> PhD Advisor	University of Massachusetts, Amherst <i>pdatta@umass.edu</i>
Dr. Anindya Iqbal <i>Professor, Computer Science and Engineering</i> Undergrad Thesis Advisor	Bangladesh University of Engineering and Technology <i>anindya@cse.buet.ac.bd</i>